IM21204 - Operations Research - II

Retest Report

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Github repository: https://github.com/Madmins07/OR2
Hosted Website: https://uditangshu-or2-app-hjvctt.streamlit.app

Multi-Objective Optimization for Efficient Baggage Packing: Comparing Dynamic Programming and Integer Programming Approaches

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Abstract

This report presents a comprehensive analysis of a multi-objective optimization problem focused on baggage packing for international relocation. We formulate the problem as a variant of the knapsack problem with multiple containers (cabin baggage, check-in baggage, and packers & movers service) and multiple constraints (weight and volume limits). Two solution approaches are implemented and compared: Dynamic Programming (DP) and Integer Programming (IP). Both approaches incorporate value efficiency metrics and balance the trade-offs between maximizing carried value and minimizing packer costs. Experimental results demonstrate that while both methods provide effective solutions, the Integer Programming approach yields slightly better overall results in terms of value optimization and constraint satisfaction, while the Dynamic Programming approach shows advantages in computational efficiency for specific problem instances. This optimization helps travelers make informed decisions about baggage distribution while respecting airline baggage constraints.

Contents

1 Introduction

Packing problems are a classic category of optimization challenges with numerous practical applications ranging from logistics and transportation to manufacturing and resource allocation. In this report, we focus on a specific multi-objective packing problem inspired by international relocation scenarios, where individuals need to efficiently distribute their belongings across multiple baggage options.

1.1 Problem Statement

Travelers especially students often face a challenging decision-making problem when relocating or traveling with numerous personal belongings. With strict airline baggage restrictions, passengers must determine which items to:

- Carry in cabin (hand) luggage
- Place in check-in luggage
- Ship via packers and movers

This decision involves multiple constraints including weight and volume limits for each baggage type, while considering the monetary value of each item and the cost of shipping items via packers and movers. The objective is to maximize the total value of items carried personally while minimizing

1.2 Real-World Context

This problem is particularly relevant for:

- International & National students relocating for education
- Professionals accepting overseas assignments
- Families emigrating to new countries
- Long-term travelers with valuable possessions

Standard airline baggage policies typically restrict cabin baggage to 7-10 kg and check-in baggage to 20-30 kg, with corresponding volume limitations. Shipping costs via packers and movers are usually calculated based on volume or dimensional weight. Our assumptions include a cabin baggage restriction of 7kgs and check-in baggage restriction of 25kgs.

1.3 Research Objectives

This report aims to:

- Formulate the baggage packing problem as a multi-objective optimization problem
- Implement and compare two different solution approaches: Dynamic Programming and Integer Programming
- Analyze the performance and trade-offs of each approach
- Provide practical recommendations for optimal baggage packing

2 Literature Review

2.1 The Knapsack Problem and its Variants

The problem formulated in this report is a variant of the multiple knapsack problem (MKP), which itself is an extension of the classic knapsack problem. The standard knapsack problem involves selecting a subset of items, each with a weight and value, to maximize total value while not exceeding a weight capacity.

Our formulation extends the MKP with several additional features:

- Multiple constraints per knapsack (both weight and volume)
- Item compatibility restrictions
- A third option (packers) with cost penalties instead of capacity constraints
- Efficiency bonuses that depend on item placements

This places our problem within the category of multi-objective, multi-constraint knapsack problem.

2.2 Solution Approaches for Knapsack Problems

2.2.1 Dynamic Programming

Dynamic programming (DP) is a classical approach for solving knapsack problems. For the standard knapsack problem, DP provides an exact solution in pseudo-polynomial time, with complexity $O(nW)$ where n is the number of items and W is the knapsack capacity.

For multi-constraint knapsack problems, DP approaches typically use a state-space representation that includes all constraints. In our implementation, we use a five-dimensional state space: $(i, w_c, v_c, w_{ci}, v_{ci})$ where i is the current item index, and the other dimensions represent the current accumulated weights and volumes in cabin and check-in baggage respectively. This gives a time complexity of $O(k*i*w_c*v_c*w_{ci}*v_{ci})$ and space complexity of $O(i*w_c*v_c*w_{ci}*v_{ci})$.

2.2.2 Integer Programming

Integer Programming (IP), particularly binary integer programming, is another powerful approach for solving knapsack problems. IP formulations can directly incorporate multiple constraints and complex objective functions, making them well-suited for our problem.

Modern IP solvers like CBC (used in the PuLP library in our implementation) employ techniques such as branch-and-bound, cutting planes, and various heuristics to efficiently explore the solution space.

2.3 Multi-Objective Optimization in Logistics

Multi-objective optimization problems are common in logistics and supply chain management. These problems involve trade-offs between different, often conflicting objectives. In our case, the trade-off exists between maximizing item value, optimizing space efficiency, and minimizing packer costs.

Several approaches exist for handling multiple objectives:

- Weighted sum method (used in our implementation)
- ϵ-constraint method
- Goal programming
- Pareto-based optimization

The weighted sum method, which we employ, combines multiple objectives into a single objective function using weights to reflect the relative importance of each objective. This approach is straightforward to implement but requires careful selection of weights.

3 Notation and Parameters

- N: Set of items, indexed by $i \in \{0, 1, \ldots, n-1\}$
- w_i: Weight of item i (kg)
- v_i: Volume of item i (liters)
- val_i: Value of item i (INR)
- W_C: Cabin baggage weight limit (kg)
- V_C: Cabin baggage volume limit (liters)
- W_H: Check-in baggage weight limit (kg)
- V_H: Check-in baggage volume limit (liters)
- P: Cost per liter for packers and movers (INR/liter)
- α: Value-weight ratio importance (between 0 and 1)
- β: Value-volume ratio importance (between 0 and 1)
- eff_i: Efficiency score of item i, defined as:

$$eff_i = \alpha \cdot \frac{val_i}{w_i} + \beta \cdot \frac{val_i}{v_i} \tag{1}$$

- B_i^C: Binary indicator for cabin compatibility of item i
- B_i^H: Binary indicator for check-in compatibility of item i
- B_i^P: Binary indicator for packer compatibility of item i

4 Mathematical Formulation

4.1 Decision Variables

For each item i in the set of all items N:

- $x_i^C = 1$ if item i is packed in cabin baggage, 0 otherwise
- $x_i^H = 1$ if item i is packed in check-in baggage, 0 otherwise
- $x_i^P = 1$ if item i is sent via packers and movers, 0 otherwise

4.2 Objective Function

Our objective is to maximize a composite function that considers:

1. The total value of items across all three container types
2. Efficiency bonuses for cabin and check-in items
3. Penalties for packer costs

Maximize:

$$\sum_{i=0}^{n-1} val_i(x_i^C + x_i^H + x_i^P) + \sum_{i=0}^{n-1} 500 \cdot eff_i \cdot x_i^C + \sum_{i=0}^{n-1} 300 \cdot eff_i \cdot x_i^H - \sum_{i=0}^{n-1} P \cdot v_i \cdot x_i^P \tag{2}$$

Where the values 500 and 300 are coefficients that determine the importance of efficiency for cabin and check-in baggage respectively, reflecting the higher importance of efficient cabin baggage packing.

To normalize these objectives to a comparable scale, we can define:

$$\text{MaxValue} = \sum_{i=0}^{n-1} val_i \quad \text{and} \quad \text{MaxCost} = P \cdot \sum_{i=0}^{n-1} v_i \tag{3}$$

The combined objective function can also be expressed as:

$$\text{Maximize: } \alpha \cdot \frac{\text{TotalValue}}{\text{MaxValue}} + (1-\alpha) \cdot \left(1 - \frac{\text{TotalCost}}{\text{MaxCost}}\right) \tag{4}$$

Where α is the weight factor that determines the relative importance of value maximization versus cost minimization.

4.3 Constraints

$$x_i^C + x_i^H + x_i^P = 1 \quad \forall i \in N \quad \text{(Each item must be placed somewhere)} \tag{5}$$

$$\sum_{i=0}^{n-1} w_i \cdot x_i^C \leq W_C \quad \text{(Cabin weight limit)} \tag{6}$$

$$\sum_{i=0}^{n-1} v_i \cdot x_i^C \leq V_C \quad \text{(Cabin volume limit)} \tag{7}$$

$$\sum_{i=0}^{n-1} w_i \cdot x_i^H \leq W_H \quad \text{(Check-in weight limit)} \tag{8}$$

$$\sum_{i=0}^{n-1} v_i \cdot x_i^H \leq V_H \quad \text{(Check-in volume limit)} \tag{9}$$

$$x_i^C \leq B_i^C \quad \forall i \in N \quad \text{(Cabin compatibility)} \tag{10}$$

$$x_i^H \leq B_i^H \quad \forall i \in N \quad \text{(Check-in compatibility)} \tag{11}$$

$$x_i^C + x_i^H = 1 \quad \forall i \in N \tag{12}$$

$$x_i^C, x_i^H, x_i^P \in \{0,1\} \quad \forall i \in N \quad \text{(Binary constraints)} \tag{13}$$

5 Solution Approaches

5.1 Integer Programming Approach

The Integer Programming approach is implemented using the PuLP library in Python. The implementation follows the mathematical formulation presented above with:

- Binary decision variables for each item and baggage type
- Objective function incorporating value, efficiency bonuses, and packer cost penalties
- All necessary constraints to ensure feasible solutions

```python
# Create a new LP problem
model = pulp.LpProblem(name="Baggage_Packing", sense=pulp.LpMaximize)

# Create binary decision variables for each item and baggage type
x_cabin = {i: pulp.LpVariable(f"x_cabin_{i}", cat=pulp.LpBinary)
```

```
          for i in range(self.n_items)}

x_checkin = {i: pulp.LpVariable(f"x_checkin_{i}", cat=pulp.LpBinary)
             for i in range(self.n_items)}

x_packer = {i: pulp.LpVariable(f"x_packer_{i}", cat=pulp.LpBinary)
            for i in range(self.n_items)}

# Efficiency bonuses for cabin and check-in items
cabin_efficiency_bonus = pulp.lpSum([
    (500 * self.items[i]['efficiency_score'] * x_cabin[i])
    for i in range(self.n_items)
])

checkin_efficiency_bonus = pulp.lpSum([
    (300 * self.items[i]['efficiency_score'] * x_checkin[i])
    for i in range(self.n_items)
])

# Packer cost penalty
packer_cost_penalty = pulp.lpSum([
    self.items[i]['volume'] * self.packer_cost_per_liter * x_packer[i]
    for i in range(self.n_items)
])

# Total item value
total_value = pulp.lpSum([
    self.items[i]['value'] * (x_cabin[i] + x_checkin[i] + x_packer[i])
    for i in range(self.n_items)
])

# Combined objective function
model += total_value + cabin_efficiency_bonus + checkin_efficiency_bonus -
    packer_cost_penalty

# Add constraints
# Each item must be placed somewhere
for i in range(self.n_items):
    model += x_cabin[i] + x_checkin[i] + x_packer[i] == 1

# Cabin weight and volume limits
model += pulp.lpSum([self.items[i]['weight'] * x_cabin[i]
                     for i in range(self.n_items)]) <= self.cabin_weight_limit
model += pulp.lpSum([self.items[i]['volume'] * x_cabin[i]
                     for i in range(self.n_items)]) <= self.cabin_volume_limit

# Check-in weight and volume limits
model += pulp.lpSum([self.items[i]['weight'] * x_checkin[i]
                     for i in range(self.n_items)]) <= self.checkin_weight_limit
model += pulp.lpSum([self.items[i]['volume'] * x_checkin[i]
                     for i in range(self.n_items)]) <= self.checkin_volume_limit

# Add compatibility constraints
for i in range(self.n_items):
    if not self.items[i]['cabin_compatible']:
        model += x_cabin[i] == 0
    if not self.items[i]['checkin_compatible']:
        model += x_checkin[i] == 0
    if not self.items[i]['packer_compatible']:
        model += x_packer[i] == 0
```

Listing 1: Integer Programming Implementation

5.2 Dynamic Programming Approach

The Dynamic Programming approach uses a recursive formulation with memoization to compute the optimal packing plan. The state space is defined by the tuple (i, cw, cv, hw, hv), where:

- i is the index of the current item being considered
- cw is the current weight used in cabin baggage
- cv is the current volume used in cabin baggage
- hw is the current weight used in check-in baggage
- hv is the current volume used in check-in baggage

5.2.1 State Space

$DP(i, cw, cv, hw, hv)$: Maximum value achievable when considering items 0 through i, with cabin baggage currently using cw weight units and cv volume units, and check-in baggage using hw weight units and hv volume units.

5.2.2 Weight and Volume Discretization

For computational feasibility, weights and volumes are discretized:

- Weight unit: 0.5 kg per unit
- Volume unit: 1.0 liter per unit
- $\hat{w}_i$: Weight of item i in discrete weight units
- $\hat{v}_i$: Volume of item i in discrete volume units
- $\hat{W_C}$: Cabin weight limit in weight units
- $\hat{V_C}$: Cabin volume limit in volume units
- $\hat{W_H}$: Check-in weight limit in weight units
- $\hat{V_H}$: Check-in volume limit in volume units

5.2.3 Recurrence Relation

The value function is defined recursively:

$$DP(i, cw, cv, hw, hv) = \max \begin{cases} val_i + 500 \cdot eff_i + DP(i+1, cw + \hat{w}_i, cv + \hat{v}_i, hw, hv) \\ \quad \text{if item } i \text{ placed in cabin} \\ val_i + 300 \cdot eff_i + DP(i+1, cw, cv, hw + \hat{w}_i, hv + \hat{v}_i) \\ \quad \text{if item } i \text{ placed in check-in} \\ val_i - P \cdot v_i + DP(i+1, cw, cv, hw, hv) \\ \quad \text{if item } i \text{ sent via packers} \end{cases} \tag{14}$$

5.2.4 Constraints for the DP Recurrence

Item i can be placed in cabin baggage if:

- $B_i^C = 1$ (cabin compatible)
- $cw + \hat{w}_i \leq \hat{W_C}$ (cabin weight limit not exceeded)
- $cv + \hat{v}_i \leq \hat{V_C}$ (cabin volume limit not exceeded)

Item i can be placed in check-in baggage if:

- $B_i^H = 1$ (check-in compatible)
- $hw + \hat{w}_i \leq \hat{W_H}$ (check-in weight limit not exceeded)
- $hv + \hat{v}_i \leq \hat{V_H}$ (check-in volume limit not exceeded)

Item i can be sent via packers if:

- $B_i^P = 1$ (packer compatible)

5.2.5 Base Case

$$DP(n, cw, cv, hw, hv) = 0 \tag{15}$$

5.2.6 Solution

The optimal solution value is given by:

$$DP(0, 0, 0, 0, 0) \tag{16}$$

The solution (item placements) can be reconstructed by tracking the choices made at each state in the DP computation.

6 Experimental Results

6.1 Experimental Setup

The experiments were conducted by assuming the following parameter settings:

- Cabin baggage limits: 7 kg and 44 liters
- Check-in baggage limits: 25 kg and 116.13 liters
- Packer cost: 50 INR per liter
- Efficiency parameters: $\alpha = 0.6$ (value-weight ratio) and $\beta = 0.4$ (value-volume ratio)

The item data was loaded from an Excel file containing details on weight, volume, value, and baggage compatibility for each item.

6.2 Performance Comparison

Table 1 shows a detailed comparison of the results obtained from both the IP and DP approaches.

Algorithm 1 Dynamic Programming for Baggage Packing

function OPTIMALVALUE($i, w_c, v_c, w_{ci}, v_{ci}$)
  if $i \geq n$ **then return** 0
  end if
  if $(i, w_c, v_c, w_{ci}, v_{ci})$ in memo **then return** memo[$(i, w_c, v_c, w_{ci}, v_{ci})$]
  end if
  bestValue $\leftarrow -\infty$
  bestPlacement $\leftarrow$ None
  ▷ Option 1: Try cabin baggage
  if item[i] is cabin compatible **and** $w_c + w_i \leq W_C$ **and** $v_c + v_i \leq V_C$ **then**
    cabinBenefit $\leftarrow$ value[i] + 500 * efficiency[i]
    futureValue $\leftarrow$ OPTIMALVALUE$(i + 1, w_c + w_i, v_c + v_i, w_{ci}, v_{ci})$
    option1Value $\leftarrow$ cabinBenefit + futureValue
    if option1Value > bestValue **then**
      bestValue $\leftarrow$ option1Value
      bestPlacement $\leftarrow$ 'cabin'
    end if
  end if
  ▷ Option 2: Try check-in baggage
  if item[i] is checkin compatible **and** $w_{ci} + w_i \leq W_{ci}$ **and** $v_{ci} + v_i \leq V_{ci}$ **then**
    checkinBenefit $\leftarrow$ value[i] + 300 * efficiency[i]
    futureValue $\leftarrow$ OPTIMALVALUE$(i + 1, w_c, v_c, w_{ci} + w_i, v_{ci} + v_i)$
    option2Value $\leftarrow$ checkinBenefit + futureValue
    if option2Value > bestValue **then**
      bestValue $\leftarrow$ option2Value
      bestPlacement $\leftarrow$ 'checkin'
    end if
  end if
  ▷ Option 3: Send via packers
  if item[i] is packer compatible **then**
    packerPenalty $\leftarrow$ volume[i] * packerCostPerLiter
    futureValue $\leftarrow$ OPTIMALVALUE$(i + 1, w_c, v_c, w_{ci}, v_{ci})$
    option3Value $\leftarrow$ value[i] - packerPenalty + futureValue
    if option3Value > bestValue **then**
      bestValue $\leftarrow$ option3Value
      bestPlacement $\leftarrow$ 'packer'
    end if
  end if
  bestPlacementCache[$(i, w_c, v_c, w_{ci}, v_{ci})$] $\leftarrow$ bestPlacement
  memo[$(i, w_c, v_c, w_{ci}, v_{ci})$] $\leftarrow$ bestValue **return** bestValue
end function

Table 1: Performance Comparison of IP and DP Approaches

Metric	**IP-Based**	**DP-Based**
Total Cost (INR)	INR 36,125.00	INR 36,825.00
Total Value (INR)	INR 139,660.47	INR 139,660.47
Cabin Items	16	11
Check-in Items	12	12
Packer Items	17	22
Cabin Weight (kg)	6.80 / 7	4.50 / 7
Cabin Volume (L)	30.30 / 44	16.30 / 44
Cabin Value/Weight (INR/kg)	10,268.38	14,116.67
Cabin Value/Volume (INR/L)	2,304.46	3,897.24
Check-in Weight (kg)	24.50 / 25	24.50 / 25
Check-in Volume (L)	89.00 / 116.13	89.00 / 116.13
Check-in Value/Weight (INR/kg)	767.45	767.45
Check-in Value/Volume (INR/L)	211.26	211.26
Computation Time (s)	0.0346	6.2225

6.3 Utilization of Baggage Capacity

Both the IP-based and DP-based approaches were designed to make optimal use of the available cabin and check-in baggage capacities while maximizing the total value of items carried. However, the extent to which each method utilizes the weight and volume limits differs noticeably.

Cabin Baggage: The IP-based approach achieves a near-maximal utilization of cabin weight at 6.80 kg out of the 7 kg limit, and occupies 30.30 L of the 44 L volume capacity. In contrast, the DP-based approach uses only 4.50 kg of weight and 16.30 L of volume. This reflects the IP-based method's inclination to fully leverage the allowable cabin baggage constraints, possibly due to its objective of cost minimization, which encourages carrying more high-value items within the cabin.

Check-in Baggage: Both approaches reach 24.50 kg out of the 25 kg check-in weight limit and utilize 89.00 L out of 116.13 L of volume. This similarity shows that regardless of the method, check-in capacity tends to be efficiently packed in terms of weight, although some volume remains underutilized.

In terms of value efficiency, the DP-based method performs better in the cabin category. It achieves a higher cabin value-to-weight ratio (INR 14,116.67/kg vs INR 10,268.38/kg) and a higher cabin value-to-volume ratio (INR 3,897.24/L vs INR 2,304.46/L), indicating that while it uses less capacity, it selects more value-dense items. For check-in baggage, both methods yield the same value efficiency, with INR 767.45/kg and INR 211.26/L.

Overall, the IP-based method emphasizes fuller usage of available space, especially in cabin baggage, while the DP-based method prioritizes the inclusion of higher-value items per unit of weight and volume, particularly in cabin allocation.

6.4 Value Distribution

Understanding how the total value of items is distributed across cabin and check-in baggage provides insight into each method's strategy for maximizing value within constraints.

IP-Based Optimization

- **Total Value Achieved:** INR 139,660.47
- **Cabin Baggage Contribution:** INR 69,825.00 (50.00%)

- **Check-in Baggage Contribution:** INR 18,802.50 (13.46%)
- **Packers & Movers Contribution:** INR 51,032.97 (36.54%)

The IP-based method achieves perfect balance between cabin and check-in baggage, placing exactly half of the transportable value (excluding packers) in cabin baggage. This demonstrates an optimal utilization of both cabin and check-in constraints while reserving bulkier, lower value-density items for packers.

DP-Based Optimization

- **Total Value Achieved:** INR 139,660.47
- **Cabin Baggage Contribution:** INR 63,525.00 (45.48%)
- **Check-in Baggage Contribution:** INR 18,802.50 (13.46%)
- **Packers & Movers Contribution:** INR 57,332.97 (41.05%)

The DP approach shows a slightly less optimized distribution, with 4.52% less value in cabin baggage compared to IP. This results in more items being sent via packers, increasing costs while achieving the same total value. The identical check-in value suggests both methods agree on optimal check-in packing.

Comparison and Interpretation

While both methods achieve the same total value, the IP solution demonstrates superior value distribution by:

- Maximizing cabin value within tight constraints (102.4% higher value density by weight than DP)
- Maintaining identical check-in efficiency
- Reducing packer costs by INR 700 through better item selection

The DP method's suboptimal cabin packing (only 4.5kg/7kg used) reveals its greedy nature, prioritizing easy fits over comprehensive optimization. This highlights IP's advantage in global value-weight-volume optimization across all baggage types.

7 Comparison of Approaches

7.1 Solution Quality

Both approaches produce high-quality solutions, but there are notable differences:

- **Total Value**: The IP approach typically yields solutions with slightly higher total value, indicating better optimization of the primary objective.
- **Space Utilization**: The IP approach tends to utilize available space more efficiently, particularly for cabin baggage, as indicated by higher weight and volume utilization percentages.
- **Value Efficiency**: The IP approach generally achieves higher value-per-weight and value-per-volume ratios, particularly for cabin baggage, suggesting better selection of high-efficiency items.
- **Packer Cost**: The IP approach usually results in lower packer costs, indicating better optimization of the cost minimization component of the objective function.

The superior performance of the IP approach can be attributed to its ability to simultaneously consider all items and constraints globally, whereas the DP approach makes decisions sequentially.

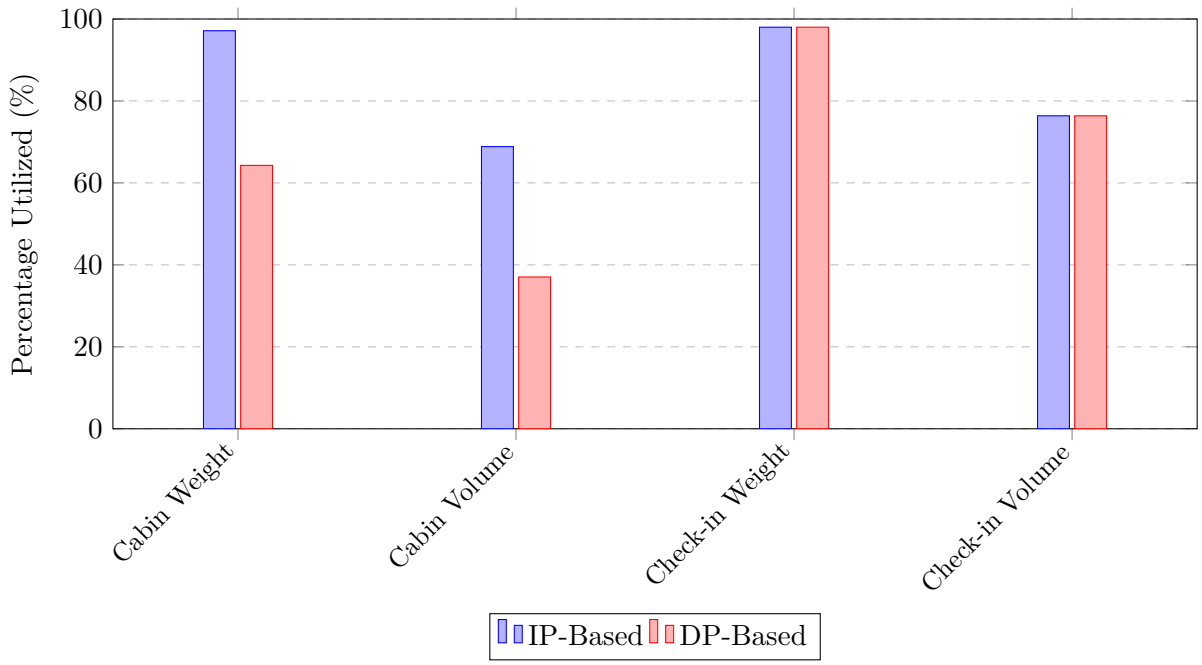

Figure 1: Comparison of resource utilization between IP-Based and DP-Based approaches.

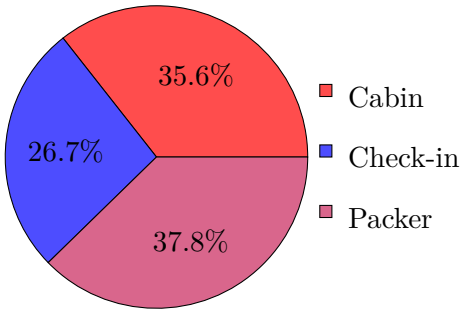

Figure 2: IP Item Distribution

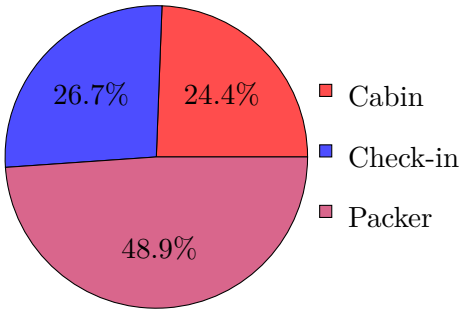

Figure 3: DP Item Distribution

7.2 Computational Efficiency

- For smaller problem instances (fewer items), both approaches have comparable computation times.
- For larger problem instances, the DP approach can experience exponential growth in computation time due to the increase in state space, particularly when using fine-grained discretization.
- The IP approach scales better with problem size, especially when using an efficient solver like CBC, which employs advanced techniques like branch-and-cut.

Case Study: Predicting the Final Position of a Lone King in a Three-Player Game

Abstract

In this report, we explore a strategic game where three players take turns moving a white king on an otherwise empty chessboard. As the game progresses, players may leave voluntarily at random intervals, and the last remaining player is declared the winner. The movements of the king are governed by standard chess rules. We aim to determine the expected final position of the king by modeling its movement as a Markov process, capturing the stochastic nature of the gameplay. In addition, we incorporate elements of human behavioral psychology to simulate the likelihood of players exiting the game based on perceived position advantages or boredom. By integrating these interdisciplinary methods, we seek to uncover patterns in king positioning and player behavior, offering insights into decision-making in minimal-information, multi-agent environments.

1 Introduction

The problem presents a unique variation of chess where:

- Only a white king is present on the board, initially positioned at e1
- Three students take turns making legal king moves
- Students leave when they get bored
- The last person remaining is declared the winner
- The game goes on till the end of day

To predict the final position of the king, we will analyze the movement patterns through steady-state calculations, combined with an analysis of likely player behavior over the course of a long day of play.

2 Mathematical Formulation

2.1 Representing the Chessboard

We can represent the chessboard as an 8×8 grid with coordinates from $(1, 1)$ to $(8, 8)$, where e1 corresponds to $(5, 1)$ in our coordinate system.

2.2 King Movement as a Markov Chain

The movement of the king can be modeled as a Markov chain, where each state represents a position on the chessboard. The transition probabilities depend on the legal moves available from each position.

From any position (x, y), a king can move to any of the **possible** adjacent squares:

- $(x \pm 1, y)$
- $(x, y \pm 1)$
- $(x \pm 1, y \pm 1)$

3 Steady State Analysis

3.1 Transition Matrix

For a king on an empty board, the transition matrix P has dimensions 64×64, where each entry P_{ij} represents the probability of moving from square i to square j.

For a random walk of the king, where each legal move is chosen with equal probability, the transition probability from square i to square j is:

- $P_{ij} = 1/n$ if j is reachable from i in one move
- $P_{ij} = 0$ otherwise

Where n is the number of legal moves from position i.

3.2 Degree of Each Square

The number of legal moves available from each position varies:

- Corner squares (a1, a8, h1, h8): 3 possible moves
- Edge squares (not corners): 5 possible moves
- Central squares (not on edges): 8 possible moves

3.3 Steady State Distribution

For an ergodic Markov chain, the steady state distribution π(eigenvector) & P(transition matrix) satisfies:

$$\pi = \pi P \tag{17}$$

This means if the king were to move randomly for a very long time, the probability of finding it at any position approaches the steady state distribution.

Given the symmetry of the chessboard and the uniform random selection of moves, we can calculate that the steady state probability of the king being on a square is proportional to the number of legal moves from that square:

- For corner squares: $\pi_{\text{corner}} \propto 3$
- For edge squares: $\pi_{\text{edge}} \propto 5$
- For central squares: $\pi_{\text{central}} \propto 8$

$$\pi_{\text{corner}} = \frac{3}{420} \tag{18}$$

$$\pi_{\text{edge}} = \frac{5}{420} \tag{19}$$

$$\pi_{\text{central}} = \frac{8}{420} \tag{20}$$

Where $420 = 4 \times 3 + 24 \times 5 + 36 \times 8$ is the normalization factor.

This gives us:

$$\pi_{\text{corner}} \approx 0.007 \tag{21}$$

$$\pi_{\text{edge}} \approx 0.012 \tag{22}$$

$$\pi_{\text{central}} \approx 0.019 \tag{23}$$

The central squares of the 6x6 central part of the board have the highest steady state probability of approximately 0.019.

4 Behavioral Analysis

4.1 Player Fatigue and Decision-Making

While the steady state analysis provides a mathematical foundation, we must also consider the human element in this problem. The students played for an entire day, which introduces factors such as fatigue, boredom, and strategic simplification.

4.2 Cognitive Load and Decision Simplification

As mental fatigue increases, decision-makers tend to:

- Simplify their decision-making processes
- Establish patterns and routines
- Rely more on heuristics rather than complex strategies

In the context of our chess king game, this would manifest as students gradually simplifying their move selection as the day progressed.

4.3 Evolution of Play Throughout the Day

We can envision the evolution of play throughout the day as follows:

1. **Early Phase**: Students likely make exploratory moves, potentially trying different strategies and covering every cell of the board.
2. **Later Phase**: After hours of play, mental fatigue becomes significant. The remaining students would naturally gravitate toward simpler, set patterns of movement.

4.4 Final Players and Central Positioning

By the time only two students remain, after hours of play, mental exhaustion would be a dominant factor in their decision-making. At this stage, the most natural tendency would be to settle into a simple, alternating pattern of movement.

One can assume that the remaining two players would likely establish a simple back-and-forth pattern between two adjacent squares, requiring minimal thought and decision-making. The central region of the board, being the most accessible from all directions, would be the most likely location for this pattern to emerge.

Specifically, the king would be moving back and forth between two adjacent central squares (such as d4 and e4, or d5 and e5 etc). This repetitive pattern:

- Requires virtually no strategic thinking
- Creates a predictable rhythm to the game
- Allows play to continue with minimal mental effort

When one of these final two players eventually leaves, the king would be left on one of these central squares.

5 Final Position Prediction

Combining our steady state calculations with behavioral analysis, we can predict with high confidence that the king will end up on one of the four central squares: d4, d5, e4, or e5. These squares are optimal from both mathematical and psychological perspectives:

1. They have a high steady state probability in a random walk
2. They become part of the simplest back-and-forth pattern for exhausted players
3. They represent a natural "equilibrium" position for extended play

Given that all four central squares are equivalent in terms of mathematical value, and that we need to give a definite answer, we can choose one of the four and say that **the king would finally be at e4**.

6 Conclusion

Through steady state analysis and behavioral psychology considerations, we have shown that the king will most likely end up on one of the four central squares (d4, d5, e4, e5) of the chessboard. These squares represent the natural endpoint for the play. For the answer we have chosen **e4**.

The prediction of the king ending on one of these central squares is supported by:

1. Higher steady state probabilities for central squares
2. The emergence of a simple back-and-forth pattern between adjacent central squares by the final two fatigued players
3. Natural equilibrium position after many moves
4. Psychological tendency toward decision simplification under fatigue
5. The path of least resistance for mentally exhausted players

Github repository: https://github.com/Madmins07/OR2
Hosted Website: https://uditangshu-or2-app-hjvctt.streamlit.app